

Robin Hur

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Education

University of Virginia, School of Engineering and Applied Science

Charlottesville, VA

B.S. in Mechanical Engineering, GPA: 3.75

Expected May 2027

- **Relevant Coursework:** Manufacturing Process Technology, Mechatronics, Strength of Materials, Statics, Dynamics, Fluid Mechanics, Thermodynamics, Materials Science, Computational Methods

Skills

- **DFM & Mechanical Design:** SOLIDWORKS (CSWA), Fusion 360, Maya, Tecplot 360, GD&T, Technical Drawings
- **Process Data Analysis & Controls:** Python, MATLAB, Simulink, Linux, ANSYS, CFD
- **Prototyping & Hardware:** Embedded Microcontrollers, Rapid Prototyping, High-Performance Computing (HPC)
- **Certifications & Languages:** English (native), Korean (fluent), French (intermediate)

Experience

Research Assistant, UVA Dept. of Materials Science and Engineering

Feb 2026 – Present

- Developed closed-loop feedback control in Python to dynamically regulate N₂ flow and relative humidity.
- Rapidly architected an error-free Python control interface via AI-assisted workflows, cross-referencing legacy humidity software and THProbe defaults to maintain relative humidity within $\pm 2\%$ over 20+ hours.
- Advanced environmental cracking test setups for high-strength aerospace aluminum alloys supporting research sponsored by NASA, Lockheed Martin, and the F-35 program.

Research Assistant, UVA Dept. of Mechanical and Aerospace Engineering

May 2026 – Present

- Executed CFD simulations and numerical design optimization on bio-inspired underwater robotic systems, evaluating part-level geometric modifications to improve propulsion efficiency and hydrodynamic performance.
- Set up automated Python and parallel processing scripts on the university HPC cluster to process 3D flow field data, accelerating iterative design reviews and thrust/drag trade studies.
- Presented weekly technical reports and design recommendation packages to cross-functional engineering teams to guide hardware iteration.

Research Assistant, UVA Dept. of Chemical Engineering

Aug 2024 – May 2026

- Analyzed atomic-level assembly mechanisms during hydrogel gelation using all-atom molecular dynamics (MD) simulations in AMBER, identifying key structural variables affecting mechanical yield.
- Built automated Python pipelines with Git version control to process simulation trajectories in VMD, managing data architecture for 10+ TB of test data across 100+ HPC configurations.

Research Output:

Co-Author, *A Methodological Approach to Relate Pentapeptide Sequence, Atomistic Self-Assembly, and Gelation Behavior*, ACS Biomaterials Science & Engineering

Jun 2026

Teaching Assistant, UVA Dept. of Electrical and Computer Engineering

Jan 2026 – May 2026

- Supported undergraduate students during embedded systems laboratories and office hours, grading 450+ students' worksheets alongside peers.
- Collaborated with faculty to evaluate technical assignments and refine lab instruction documentation for quality and clarity.

Projects

Mechatronics and Robotics Society, Mechanical Subteam Member

Aug 2025 – Dec 2025

- Modeled multi-part CAD assemblies in SOLIDWORKS for a NASA Lunabotics excavation robot, applying DFM/DFA principles to ensure modular assembly, structural rigidity, and tight-tolerance component fitment.
- Evaluated granular flow and excavation subsystem mechanisms, identifying stress concentration areas and failure modes to optimize material handling efficiency in simulated lunar soil.

Potential Energy Propulsion Racer, Mechanical Lead

Nov 2025

- Designed and fabricated a 3D-printed vehicle transmission and frame using SOLIDWORKS, conducting tolerance analysis to guarantee proper gear mesh clearances, press-fit tolerances, and low friction.
- Curated complete engineering documentation including multi-part assembly drawings, a comprehensive Bill of Materials (BOM), and mechanical trade-off analyses.